

TASK 5

Capture and Analyze Network Traffic Using Wireshark

Cybersecurity Internship Practical Report

Item	Details
Tool	Wireshark
Platform	Windows 64-bit
Activity	Live network packet capture and analysis
Protocols identified	DNS, TCP, TLS
Capture file	task5_network_capture.pcapng
Status	Completed

Purpose

This report documents the practical work completed for Task 5. Wireshark was used to capture live network traffic, filter packets by protocol, inspect packet details, identify at least three protocols, and save the capture for later analysis.

1. Objective

The objective was to capture live network packets and identify basic protocols and traffic types. The internship task requires installing Wireshark, capturing traffic on the active interface, generating traffic, stopping the capture, filtering packets, identifying at least three protocols, exporting the capture, and summarizing the findings.

2. Step-by-Step Procedure

Step 1 — Install Wireshark

Wireshark was installed on the Windows computer using the normal Windows installation process.

Step 2 — Select the active interface

The Wireshark interface showing network activity was selected and opened to start packet capture.

Step 3 — Capture traffic

The capture was allowed to collect live network traffic while normal network/web activity generated packets.

Step 4 — Stop the capture

The capture was stopped after sufficient packets had been collected.

Step 5 — Filter packets

Display filters were used to isolate DNS, TCP, TLS and HTTP traffic.

Step 6 — Inspect packets

Representative DNS, TCP and TLS packets were selected and their packet details were viewed.

Step 7 — Save the capture

The capture was saved as `task5\_network\_capture.pcapng`.

3. Filters Used and What They Mean

DNS

dns

Displays DNS packets. DNS stands for Domain Name System and is used to translate domain names into IP addresses.

TCP

tcp

Displays TCP packets. TCP stands for Transmission Control Protocol and provides connection-oriented, reliable communication.

TLS

tls

Displays TLS packets. TLS stands for Transport Layer Security and protects many network connections using encryption and authentication.

HTTP

http

Displays HTTP packets when present. Modern websites commonly use HTTPS, so a capture may contain TLS traffic instead of visible HTTP.

4. Technical Terms — Beginner Friendly

Term	Definition
Wireshark	A network protocol analyzer used to capture and inspect network packets.
Packet	A small unit of data transmitted over a network.
Protocol	A set of rules used by systems to communicate.
Network interface	The connection used by a computer to communicate, such as Wi-Fi or Ethernet.
DNS	Domain Name System; translates domain names into IP addresses.
TCP	Transmission Control Protocol; provides reliable, connection-oriented communication.
TLS	Transport Layer Security; protects network communication with encryption and authentication.
HTTP	Hypertext Transfer Protocol; used for web communication.
HTTPS	HTTP protected using TLS.
IP address	A numerical address used to identify a device or network interface.
Display filter	A Wireshark expression used to show only packets matching a condition.
PCAP/PCAPNG	File formats used to store captured network packets for later analysis.

5. Protocol Findings

Protocol	Finding
DNS	DNS traffic was observed and isolated using the `dns` display filter.
TCP	TCP traffic was observed. A TCP packet was inspected to view transport-layer information such as
TLS	TLS traffic was observed and inspected. This represents protected network communication.
HTTP	The `http` filter was tested. HTTP may be absent because modern websites commonly use HTTPS

Capture Evidence

The Wireshark capture screenshot shows live packet activity collected during the practical. The saved `.pcapng` file is the primary evidence because it preserves the captured packet data.

Screenshot reference: Wireshark packet capture showing the captured network traffic.

6. Results

The practical successfully produced a packet capture and identified three required protocols: DNS, TCP and TLS. The corresponding Wireshark filters were applied and representative packets were inspected. The capture was saved in `.pcapng` format.

7. What I Learned

- How Wireshark captures live network traffic.
- How packets are displayed and inspected.
- How display filters such as `dns`, `tcp` and `tls` help isolate relevant traffic.
- The basic purpose of DNS, TCP and TLS.
- How packet captures can help with troubleshooting and protocol analysis.
- How to save a capture for later investigation.

8. Security Importance

Packet analysis is useful in cybersecurity because it helps analysts understand communication between systems, identify protocols and endpoints, troubleshoot network problems, and investigate unusual traffic. TLS protects the contents of many connections, although some connection metadata can still be visible.

9. Conclusion

Task 5 was completed by capturing live network traffic with Wireshark, filtering and inspecting protocols, identifying DNS, TCP and TLS, and saving the capture file for documentation. The exercise provided hands-on experience with packet capture, protocol analysis, filtering and basic network troubleshooting.

FINAL STATUS: TASK 5 COMPLETED ✓

10. Internship Interview Questions — Quick Answers

What is Wireshark used for?

Wireshark is used to capture and analyze network packets and protocols.

What is a packet?

A packet is a small unit of data sent across a network.

How do you filter packets in Wireshark?

Use a display filter such as ``dns``, ``tcp`` or ``tls`` in the filter bar.

What is the difference between TCP and UDP?

TCP is connection-oriented and designed for reliable delivery; UDP is connectionless and has less overhead.

What is a DNS query packet?

It is a packet used to request DNS information, commonly to resolve a domain name.

How can packet capture help troubleshooting?

It can show connection attempts, protocols and endpoints, helping identify communication problems.

What is a protocol?

A protocol is a defined set of rules used for communication between systems.

Can Wireshark decrypt encrypted traffic?

Wireshark can analyze encrypted protocols and may decrypt some traffic when the required keys or session secrets are available. Without those secrets, encrypted application contents generally cannot simply be read.